Learning Objective 7.4: I can apply the magnetic field boundary conditions to calculate magnetic fields on both sides of a material boundary.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem, $\hat{\mathbf{n}}$ points from region 2 into region 1.

1. **Deriving the conditions.** Two magnetic materials meet at a flat boundary that carries a surface current density $\mathbf{J}_s$.

   (a) Apply Ampere's law to a rectangular loop with long sides of length W along a tangent $\hat{\mathbf{t}}$, one in each region, and short sides of height $h \to 0$. Show that $H_{t1} - H_{t2} = \mathbf{J}_s \cdot \left(\hat{\mathbf{n}} \times \hat{\mathbf{t}}\right)$.

   $$\oint \mathbf{H} \cdot d\mathbf{l} = \int_a^b \mathbf{H}_1 \cdot \hat{\mathbf{t}}\, dl + \int_c^d \mathbf{H}_2 \cdot \left(-\hat{\mathbf{t}}\right) dl + \int_b^c + \int_d^a$$
   $$= H_{t1} W - H_{t2} W + 0 = I_{\text{encl}} = \mathbf{J}_s \cdot \left(\hat{\mathbf{n}} \times \hat{\mathbf{t}}\right) W$$

   The short sides have length h, so their integrals vanish as $h \to 0$, and so does any volume current through the area Wh. The loop's normal, by the right-hand rule, is $\hat{\mathbf{n}} \times \hat{\mathbf{t}}$, so only the component of $\mathbf{J}_s$ along it crosses the loop. Divide by W.

   (b) Apply Gauss's law for magnetism to a pillbox with faces of area ΔA and height $h \to 0$. Show that $B_{n1} = B_{n2}$, and explain why no source term appears.

   $$\oint \mathbf{B} \cdot d\mathbf{s} = B_{n1} \Delta A - B_{n2} \Delta A + 0 = 0$$

   The side's flux vanishes as $h \to 0$. The right-hand side is zero for every closed surface, because there is no magnetic charge to enclose.

2. **Into a magnetic material.** The plane $z = 0$ separates air, region 1 at $z > 0$, from a material with $\mu_r = 500$, region 2 at $z < 0$. No surface current flows on the boundary. Just above it, $\mathbf{H}_1 = \hat{\mathbf{x}}10 + \hat{\mathbf{z}}20$ A/m.

   (a) Find $\mathbf{H}_2$ just below the boundary.

   The normal is $\hat{\mathbf{z}}$. Tangential: $H_{2x} = 10$ A/m. Normal:

   $$\mu_0 H_{1z} = 500\mu_0 H_{2z} \implies H_{2z} = \frac{20}{500} = 0.04 \text{ A/m}$$

   $\mathbf{H}_2 =$ $\hat{\mathbf{x}}10 + \hat{\mathbf{z}}0.04$ A/m

   (b) Find $\mathbf{B}_1$ and $\mathbf{B}_2$.

   $$\mathbf{B}_1 = \mu_0 \mathbf{H}_1 = \hat{\mathbf{x}}12.6 + \hat{\mathbf{z}}25.1 \ \mu\text{T}$$
   $$\mathbf{B}_2 = 500\mu_0 \mathbf{H}_2 = \hat{\mathbf{x}}6280 + \hat{\mathbf{z}}25.1 \ \mu\text{T}$$

   $B_z =$ $25.1 \ \mu$T, both sides

   (c) Find the angle each field makes with the normal, and check the ratio of their tangents against μ_1/μ_2.

   $$\theta_1 = \tan^{-1}\frac{10}{20} = 26.6^\circ$$
   $$\theta_2 = \tan^{-1}\frac{10}{0.04} = 89.8^\circ$$
   $$\frac{\tan\theta_1}{\tan\theta_2} = \frac{0.5}{250} = 0.002 = \frac{1}{500}$$

   $\theta_1 =$ 26.6°

   $\theta_2 =$ 89.8°

   (d) In which region is $|\mathbf{H}|$ larger, and in which is $|\mathbf{B}|$ larger? Explain in one sentence each.

   $|\mathbf{H}_1| = 22.4$ A/m is larger than $|\mathbf{H}_2| = 10.0$ A/m: the tangential parts match, and the normal part of $\mathbf{H}$ is 500 times smaller in the material. $|\mathbf{B}_2| = 6.28$ mT is larger than $|\mathbf{B}_1| = 28.1 \ \mu$T: the normal parts match, and the tangential part of $\mathbf{B}$ is 500 times larger in the material.

3. **A boundary that carries current.** The plane $z = 0$ separates region 1, $z > 0$, with $\mu = \mu_0$, from region 2, $z < 0$, with $\mu = 3\mu_0$. The boundary carries $\mathbf{J}_s = \hat{\mathbf{y}}4$ A/m. Just below it, $\mathbf{H}_2 = \hat{\mathbf{x}}2 - \hat{\mathbf{y}} + \hat{\mathbf{z}}6$ A/m.

   (a) Find $\mathbf{H}_1$.

   With $\hat{\mathbf{n}} = \hat{\mathbf{z}}$, the tangential condition gives the tangential part of $\mathbf{H}_1 - \mathbf{H}_2$ as

   $$\mathbf{J}_s \times \hat{\mathbf{n}} = 4\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}4$$

   so $H_{1x} = 2 + 4 = 6$ and $H_{1y} = -1$ A/m. Normal:

   $$\mu_0 H_{1z} = 3\mu_0 \times 6 \implies H_{1z} = 18 \text{ A/m}$$

   $\mathbf{H}_1 =$ $\boxed{\hat{\mathbf{x}}6 - \hat{\mathbf{y}} + \hat{\mathbf{z}}18 \text{ A/m}}$

   (b) Check your answer by computing $\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2)$.

   $$\hat{\mathbf{z}} \times (\hat{\mathbf{x}}4 + \hat{\mathbf{z}}12) = \hat{\mathbf{y}}4 = \mathbf{J}_s$$

4. **A current sheet and an iron core.**

   (a) A sheet of current $\mathbf{J}_s = \hat{\mathbf{y}}20$ A/m lies in the plane $x = 0$ in free space. Lesson 30 found that it makes a field of magnitude $J_s/2$ on each side, parallel to the sheet and at right angles to the current. Find $\mathbf{H}$ on each side, and show that the tangential condition holds.

   Region 1 is $x > 0$, so $\hat{\mathbf{n}} = \hat{\mathbf{x}}$. The right-hand rule gives $\mathbf{H}_1 = -\hat{\mathbf{z}}10$ and $\mathbf{H}_2 = \hat{\mathbf{z}}10$ A/m.

   $$\hat{\mathbf{x}} \times (-\hat{\mathbf{z}}20) = \hat{\mathbf{y}}20 = \mathbf{J}_s$$

   (b) Inside a ferrite core with $\mu_r = 2000$, the field meets the core's surface $85°$ from the normal. Find the angle from the normal in the air just outside, and explain why a high-permeability core guides magnetic field.

   $$\tan\theta_{\text{air}} = \frac{\tan 85°}{2000} = \frac{11.4}{2000} = 0.00572 \implies \theta_{\text{air}} = 0.33°$$

   Whatever angle the field makes inside the core, short of grazing the surface, the field in the air just outside meets the surface within a fraction of a degree of the normal, so the lines of $\mathbf{B}$ in the air turn toward the core and the flux runs along the core's own path.

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